

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

September 2025 – May 2027 (Expected)

Amherst, MA

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

JavaScript, TypeScript, Python, Java, SQL, Bash, C++

Backend & APIs

Node.js, Express, Fastify, FastAPI, REST APIs, async services & queues, distributed systems, authentication, rate limiting, multi-tenant SaaS

Data & Warehousing

Apache Spark / PySpark, ETL, data warehousing concepts, Spark UI, Hadoop / MapReduce, MongoDB, PostgreSQL, DynamoDB, Redis, SQLite

Cloud & DevOps

AWS, GCP, Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux

Testing & Web

pytest, integration & regression testing, Agile / Scrum, React

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Architected full-stack data annotation platform (**React, Node.js, MongoDB**) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.
- Owned backend reliability across the eTriage stack: monitored production logs during feature releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Spark UI, SQL, Python

- Optimized an ETL and analytics pipeline over a **~1.4B-row** dataset, cutting end-to-end runtime **~25%** (17–19 min to 14.3 min) via broadcast joins, Adaptive Query Execution, skew-join handling, shuffle-partition tuning, schema-aware reads with column pruning, and year/month partitioning with Snappy compression.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks and documented a repeatable optimization workflow, reducing analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x**.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage **Python/FastAPI** backend service that turns open-ended queries into structured tables with deterministic parsers before LLM fallback and cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback for reliability, and pipeline instrumentation, applying backend best practices for a debuggable, maintainable system.

Underdogs Fitness - Production Multi-Branch Management Platform

underdogsfitness.in

MongoDB, Express, React, Node.js, Stripe, DigitalOcean | live in production 3+ years

- Built and deployed a production gym-management platform on the **MERN** stack solo end-to-end, auth, membership, attendance, **Stripe + manual-cash hybrid payments** with a partial-payment ledger, and cron-based renewal automation; designed **multi-branch** from day 1 (every entity scoped by `branchId`, JWT-claim auth), onboarding a 2nd branch with **zero schema migration**.
- Set up **git-webhook auto-deploy** (GitHub webhook to DigitalOcean droplet on push) with Vercel for the frontend; moved **90%+** of payments through the portal and cut the client's manual workload **~70%**.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.